MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6SN7WGTA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification sheet and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, medium Mu

Outline --- 9-11 or 9-41 (EIA)

Base --- B8-6, B8-46, or B8-251, phenolic

Envelope --- T9

Cathode --- Coated unipotential

Base connections:

| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Element | --- | 2g | 2a | 2k | 1g | 1a | 1k | h | h |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec | Pp/a | Rg1 | Ehk | Ib/a | Alt |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | W | Meg | V | mA dc | ft |
| Maximum: | 6.6 | 330 | --- | --- (see note 1) | 1.1 | 250 | 20.0 | 60,000 |
| Minimum. | 6.0 | --- | --- | --- | --- | --- | 0.5 | --- |
| TEST CONDITIONS: | 6.3 | 250 | -8.0 | --- | --- | 0 | --- | --- |

GENERAL:

Qualification - Not required



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSP LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | | |
| 1301 | Heater current | | | 0.4 | II | If | 550 | 650 | mA |
| 1336 | Heater-cathode leakage | 2 | Ehk = ±250 V dc | 0.4 | II | Ihk | --- | 25.0 | µA dc |
| 1266 | Total grid current | 3,4 | | 0.4 | II | Ic | 0 | -1.5 | µA dc |
| 1256 | Electrode current (1) (anode) | 2,4 | | 0.4 | II | Ib | 5.5 | 12.5 | mA dc |
| 1256 | Electrode current (2) (anode) | 2 | Ec = -21 V dc | 0.4 | II | Ib | --- | 50 | µA dc |
| 1306 | Transconductance (1) | 2 | | 0.4 | II | Sm | 2,075 | 3,125 | µmhos |
| 1201 | Short and discontinuity detection | | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | | |
| 1211 | Insulation of electrodes | 2 | | 2.5 | S3 | --- | --- | --- | --- |
| 1306 | Transconductance (2) | 2 | Ef = 5.7 V | 2.5 | I | ΔSm Ef | --- | 15 | % |
| 1266 | Grid emission | 2,5 | Ef = 7.5 V; Ec = -21 V dc; Rg/g = 1.0 Meg | 2.5 | I | Ic | 0 | -1.5 | µA dc |
| 1256 | Electrode current (3) (anode) | 2 | Ec = -14 V dc | 2.5 | I | Ib | 20 | --- | µA dc |
| 1246 | Audio frequency noise | 3,6 | Esig = 70 mV ac; Rp = 0.01 Meg | 2.5 | I | --- | --- | --- | --- |
| 1306 | Transconductance (3) | 2 | Eb = 90 V dc; Ec = 0 | --- | --- | Sm | 2,400 | 3,600 | µmhos |
| 1316 | Amplification factor | 2 | | 2.5 | I | Mu | 18 | 23 | --- |
| 1331 | Direct-interelectrode capacitance | 2 | No shield | 6.5 | Code E | Cga | 3.0 | 4.0 | pF |
| | | | | | | Cin | 1.7 | 2.7 | pF |
| | | | | | | Cout | 0.50 | 1.20 | pF |
| 1002 | Barometric pressure, reduced | | Pressure = 55 ±5 mmHg; voltage = 500 V ac | 6.5 | See note 7 | --- | --- | --- | --- |
| 1031 | Low-frequency vibration | 3 | Rp = 2,000 ohms | 2.5 | I | Ep | --- | 50 | mV ac |
| 1041 | Shock | 8 | 450 G; Ehk = 100 V dc | 6.5 | See note 7 | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSP LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | MIN | MAX | |
| | Quality conformance inspection, part 2 -Continued | | | | | | | | |
| 1031 | Vibration-fatigue | | | 6.5 | See note 7 | --- | --- | --- | --- |
| --- | Shock and vibration-fatigue-test end points: | | | | | | | | |
| 1031 | Low-frequency vibration | 3 | Rp = 2,000 ohms | --- | --- | Ep | --- | 75 | mV ac |
| 1336 | Heater-cathode leakage | 2 | Ehk = ±250 V dc | --- | --- | Ihk | --- | 75 | µA dc |
| 1306 | Change in transconductance (1) of individual tubes | 2 | | --- | --- | ΔSmt | --- | 20 | % |
| 1256 | Electrode current (2) (anode) | 2 | Ec = -21 V dc | --- | --- | Ib | --- | 150 | µA dc |
| 1126 | Glass envelope strain | | | 2.5 | 1 | --- | --- | --- | --- |
| 1101 | Secureness of base, cap, or insert | | | --- | --- | --- | --- | --- | --- |
| 1111 | Base pin solder depth | | | --- | --- | --- | --- | --- | --- |
| 1105 | Permanence of marking | | | --- | --- | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1ST SAMPLE | ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 3 | | | | | | | | |
| 1506 | Heater-cycling life | 9 | Ef = 7.5 V; Eb = Ec = 0; Ehk = 135 V dc; 1 minute "on", 4 minutes "off" | --- | --- | --- | --- | --- | --- |
| 1516 | Stability life (2 hours) | 2 | Ehk = 250 V dc; Rg/g = 1.0 Meg; TA = room | --- | --- | --- | --- | --- | --- |
| --- | Stability life-test end point: | | | | | | | | |
| 1306 | Change in transconductance (1) of individual tubes | 2 | | --- | --- | $\Delta Sm_t$ | --- | 10 | % |
| 1501 | Intermittent life | 2,10 | Ehk = 250 V dc, Rg/g = 1.0 Meg | --- | --- | --- | --- | --- | --- |
| --- | Intermittent life-test end points (500 hours): | | | | | | | | |
| 1266 | Total grid current | 2 | | 1 | 3 | Ic | 0 | 1.5 | µA dc |
| 1301 | Heater current | | | 1 | 3 | If | 550 | 650 | mA |
| 1306 | Transconductance (1) | 2 | | --- | --- | $\Delta Sm_t$ | --- | 15 | % |
| 1306 | Change in transconductance (2) of individual tubes | 2 | Ef = 5.7 V | 2 | 5 | $\Delta Sm_{Ef}$ | --- | 15 | % |
| 1306 | Transconductance (1), average change | 2 | | --- | --- | $Av_\Delta Sm_t$ | --- | 15 | % |
| 1336 | Heater-cathode leakage | 2 | Ehk = ±250 V dc | 1 | 3 | Ihk | --- | 25 | µA dc |
| 1211 | Insulation of electrodes | 2 | | 2 | 5 | --- | --- | --- | --- |

NOTES:

1. The design application maximum rating of the tube shall not exceed 2.7 watts per anode power dissipation.
2. Test each unit separately.
3. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
4. This test shall be performed at the conclusion of the holding period.



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NOTES - Continued

5. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

| Ef | Ec | Eb | Rg |
|---|---|---|---|
| V | V dc | V dc | Meg |
| 7.5 | -8 | 250 | 1.0 |

6. The rejection level shall be set at the VU meter reading obtained during calibration.

7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of a lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

8. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.

9. This test shall be made on a lot-by-lot basis.

10. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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